# Prime-field index calculus: the distance to rho in the lab's own observable, the equation budget every new lead must meet, and two cheap tests

2026-11-10 · `research/prime_ic_leads_20261110/` · crypto `main` f14b66786b · desk study, derivations and a development check; no S, no cost ratio, no leaderboard row; every measurement here is `[pcc:unverified]`

## 1. Answer

On the generic track parity with rho is impossible and the gap widens like $r^{1/6}$; its value is measured constant gains. On the algebraic track the only exponent lane on record, the PKM tower family, is converging to closure: the last fall degree $d_F$ grows with the tower height at one degree per four levels (C-step holds at $t = 11, 12$; $t = 13$ reads $\geq 6$) and with the arity at half a degree per unknown bit (stage C). Section 3 turns that growth into the distance to parity in the lab's own observable; section 4 names the mechanism as an *equation budget*: one cross-summand equation per relation over $F_p$, every further one costing about $\log_2 p$ unknown bits, against one per bit for Weil descent over $F_{2^n}$. Every idea examined today fails that budget (section 5); the one family that passes it for free, the $[k]$-multiple consequences of the relation, is implemented as a presentation control with pre-stated predictions (section 6). Nothing here reaches parity and nothing on record can; the note states exactly what a new lead would have to supply.

## 2. Where the prime-field program stands (measured, by path)

| item | value | source |
|---|---|---|
| Generic track, Table B, `icv1-fp10`…`fp24` | IC / rho-negation 5.47× to 13.5× | `docs/ic/LEADERBOARD.md` Table B |
| 20-bit factor base $N = 32 \to 45$ | gap 9.47× ⟶ 7.75× [6.92, 8.70]; second-host replay 1,008/1,008 | `research/prime_base_oracle_tuning_20261029/` |
| Algebraic-track prime baseline | empty (R1 Option C) | `research/prime_ic_program_20261029/DECISIONS.md` |
| `pkm-semaev` smoke, Curve25519 miniature $k = 20$, $t = 2$ | S = 1027 / 49,513 / 485,893 at $w$ = 0 / 0.1 / 1 vs 9.05 / 9.46 / 13.16 generic | `problems/curve25519-family-algebraic-ic-tower.md` |
| PKM Kummer $m = 2$, $P_1$, $d_F$ by $t$ | 3, 4, 4, 4, 4, 5, 5, 5, 5, 6 ($t \leq 10$); 6 ($t = 11, 12$); $\geq 6$ ($t = 13, 14$) | `research/pkm_falldegree_unblock_20261107/RESULTS.md`, `RESULTS_T12_BRACKET.md` |
| Stage C, fixed $t$, growing $m$ | $t = 2$: 4, 5, 6, 6, $\geq 7$; $t = 3$: 4, 6, 7, $\geq 8$, $\geq 8$ at $m = 2..6$; "grows in m" on all six ladders | `RESULTS_STAGE_C.md` |
| Exponent study, symmetrised subgroup base | $m = 2$: $s = 2.77$ [2.31, 3.23] at $k = 20$; $m = 3, 4$ undetermined (censored) | `research/curve25519_exponent_study_20261108/` |
| Decisive cells | $t = 14$ cap 7 queued; $m = 4$, $N = 20$ closure not launched (112 GiB, 49 h, Spot) | `docs/session-logs/2026-11-10.md` |

## 3. What parity requires (derived)

With base size $B$, arity $m$, per-target cost $T = B^s$ and a sparse solve $\approx mB^2$, the total is about $r^{2/(m+1-s)}$, below $r^{1/2}$ iff $s < m - 3$ (program note §2.4; audit `semaev_alternatives_audit_20261104`). A dense closure or Macaulay elimination reaching degree $d = \alpha N$ in $N = m \log_2 B$ one-bit unknowns meets $M(N, d) = \sum_{k \leq d} \binom{N}{k} \approx 2^{NH(\alpha)}$ columns, so $s \geq mH(\alpha)\omega'$ and the lane can reach parity only if $H(\alpha) < (1 - 3/m)/\omega'$:

| m | $\alpha^*, \omega' = 1$ | $\alpha^*, \omega' = 2$ | $\alpha^*, \omega' = 3$ |
|---|---|---|---|
| 4 | 0.042 | 0.017 | 0.010 |
| 5 | 0.079 | 0.031 | 0.019 |
| 6 | 0.110 | 0.042 | 0.025 |
| 8 | 0.156 | 0.056 | 0.033 |
| 10 | 0.189 | 0.066 | 0.038 |

Measured slopes: $1/8 = 0.125$ per bit in $t$ (C-step, $m = 2$) and 0.5 per bit in $m$ (stage C, $t = 2$ and $t = 3$). Both exceed every threshold at $\omega' \geq 2$; the $m$-slope exceeds every threshold at $\omega' = 1$ too. Recorded `max_cols` = 16,572 at $t = 13$, cap 5, against $M(26, 5) = 83,682$: F4′s matrices are about 5× sparser than the dense count, a constant that does not move $s$. This is a lower

bound for dense degree-$d$ elimination and a model for F4; it assumes the slope persists beyond $N \leq 28$, which the $m = 4$, $N = 20$ closure tests.

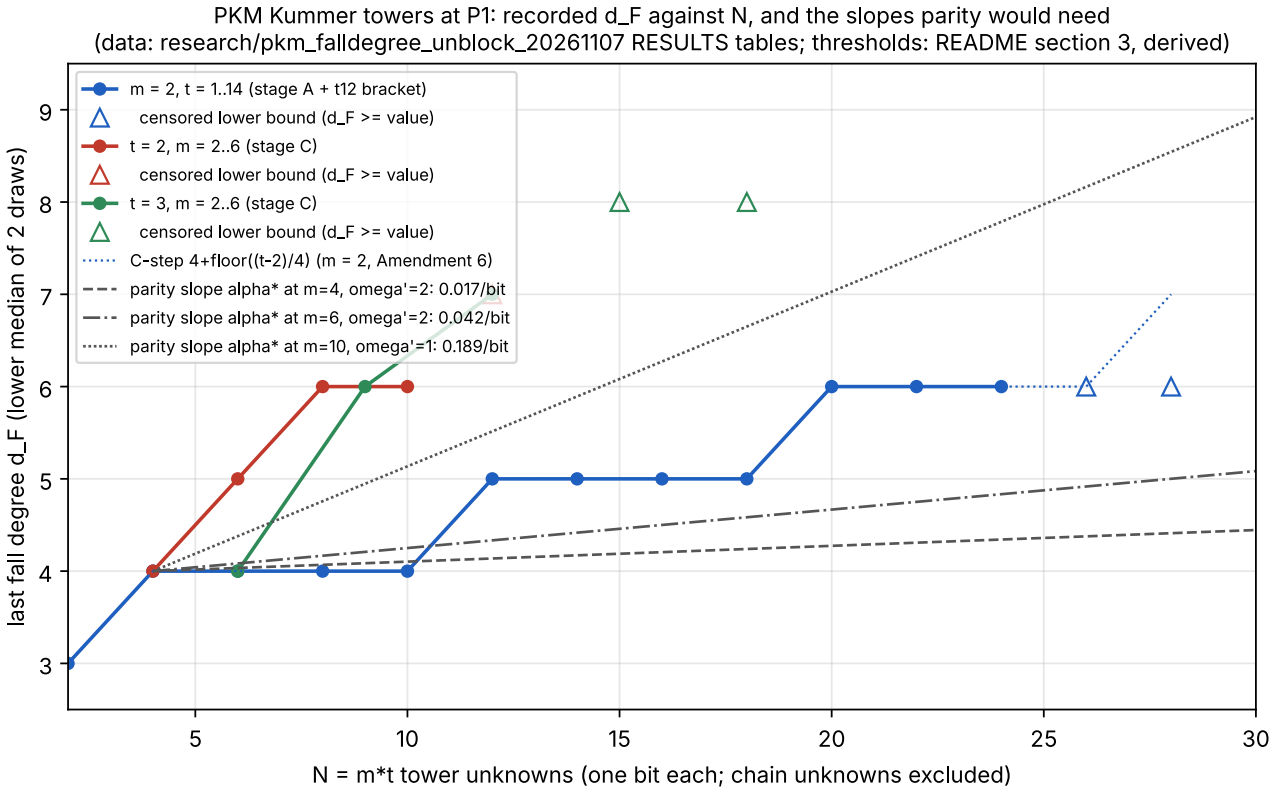



Figure 1: Recorded $d_F$ ladders (lower medians of 2 draws; triangles are censored lower bounds) against $N$, with the C-step prediction and the $\alpha^*$ parity slopes of section 3. Data: `research/pkm_falldegree_unblock_20261107` RESULTS tables; thresholds derived here.

## 4. The cross-summand equation budget (derived)

Weil descent over $F_{2^n}$: one relation gives $n$ equations of degree 2 in $mn' \approx n$ bit unknowns. A PKM tower over $F_p$: $t = \log_2 B$ one-bit unknowns per summand and **one** equation, with no Galois group to split it. A second equation from the same relation costs about $\log_2 p$ unknown bits in every route: a fixed rational image $\varphi(P)$ in its own tower ($|V| \cdot |V'| = Bp$), the extension $F_{p^k}$ (Gaudry cost $p^{2-2/k}$), a $p$-adic digit (the torsion lift's second digit has degree about $N^2$), or a jet component. The only free family is $[k]R = \sum [k]P_i$, which enters the Kummer quotient at degree about $2\log_2 k + 2$ (6 for $k = 2$). Hence a tower system is a one-generator system in a $2^N$-point quotient ring and its fall degree grows. The filter for any new lead: equations per unknown bit after the membership equations are removed; whether every extra equation brings a new $F_p$ unknown or shrinks the base; the degree at which consequence equations enter.

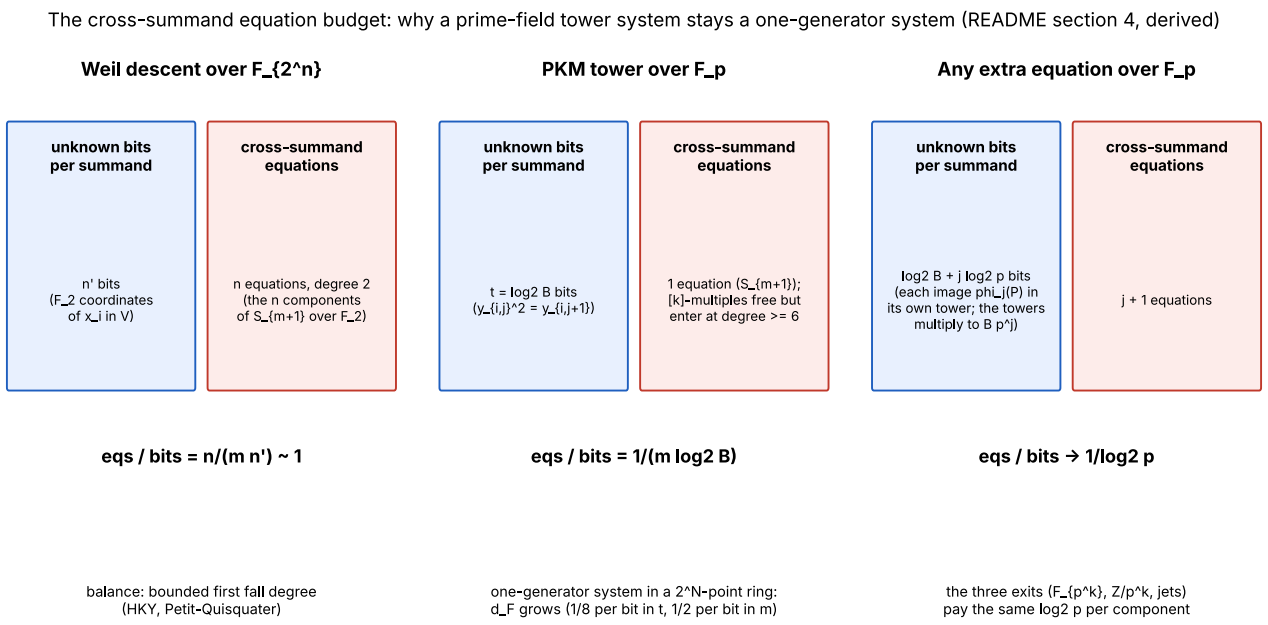



Figure 2: The equation budget: Weil descent (balance), the PKM tower (one generator), any extra equation over $F_p$ (ratio $1/\log_2 p$).

## 5. Examined today and closed on paper

2-torsion ($T_0$) symmetrised tower on Montgomery-form curves: the 2-isogeny with kernel $(0,0)$ has $x$-map $x \mapsto x + A + B/x$, so a tower on the invariant is a plain tower on $E/\langle T_0 \rangle$, same one-generator system (constant column gain only; Kummer unavailable on Curve25519, $v_2(p-1) = 2$). Translate-pair and endomorphism-image bases: $|V| \cdot |V'| = pB$, the second equation costs $\log_2 p$ bits. Wagner $k$-tree filtering on tower sets: needs $|F| \geq \sqrt{2p}$. $p$-adic second digit of the torsion lift: degree $N^2$, or a free $F_p$ unknown per summand. For $j = 0$, the $\omega$-conjugate equation is the original up to relabelling ($S_3(\zeta x) = \zeta S_3(x)$). Atlas matches are listed in the README.

## 6. The multiples presentation control

`examples/pkm_tower_pilot.rs --controls multiples` appends to the tower system the cleared $S_{m+1}(x([2]P_1), \ldots, x([2]P_m), x([2]R)) = 0$, with $x([2]P) = N(x)/D(x)$, $N = x^4 - 2ax^2 - 8bx + a^2$, $D = 4(x^3 + ax + b)$ (for chains, doubled partial-sum unknowns linked by $D(u_k)u'_k - N(u_k) = 0$). Same cell seed, same tower, curve and targets as the tower control; every appended equation vanishes at planted solutions (asserted; 12/12 unit tests). It enters at multilinear degree 6, so at $m = 2$ the two presentations must agree wherever $d_F \leq 5$ (P1, a built-in control); at cells with $d_F(\text{tower}) \geq 7$ the prediction was $d_F(\text{multiples}) \in \{d_F - 1, d_F\}$ with the slopes unchanged (P2). Falsifier: a drop of at least 2 at a cell with $d_F(\text{tower}) \geq 8$, or an $m$-ladder that reads flat.

**Development check** (this host, closure engine, Kummer at $P_1$, seed `0x504B4D54`, two random targets per cell, both arms on the same targets; no isolation; every value `[pcc:unverified]`; rows and table in `devcheck/`). The tower arm reproduces stage A's medians at $t = 2..9$ and stage C's cells, the instrument check.

| cell | $d_F$ tower | $d_F$ multiples | $\Delta$ | falls tower / multiples |
|---|---|---|---|---|
| $m = 2, t = 2..9$ (16 rows) | 4, 4, 4, 4, 5, 5, 5, 5 | identical on every row | 0 | identical |
| $m = 2, t = 10$ | > 6 (budget-censored) | > 6 (same) | — | [4, 5] / [4, 5] |
| $m = 3, t = 2$ ($N = 6$) | 5, 5 | 5, 5 | 0 | [4, 5] / [4, 5] |
| $m = 3, t = 3$ ($N = 9$) | 6, 6 | 5, 5 | **−1** | [4, 5, 6] / [4, 5] |
| $m = 4, t = 2$ ($N = 8$) | 6, 6 | 5, 5 | **−1** | [4, 5, 6] / [4, 5] |
| $m = 4, t = 3$ ($N = 12$) | 7, 7 | 6, 6 | **−1** | [4, 5, 6, 7] / [4, 5, 6] |
| $m = 5, t = 2$ ($N = 10$) | 6, 6 | 6, 6 | 0 | [4, 5, 6] / [4, 5, 6] |
| $m = 5, t = 3$ ($N = 15$) | > 8 (budget-censored, both draws) | per-draw runs, cmax 7: `devcheck/m5_t3_mult_k{0,1}.jsonl` | — | [4, 5, 6, 7] / — |

P1 holds (identical at $m = 2$, draw by draw). P2 holds and nothing reaches the falsifier: at $m \geq 3$ the drop is exactly one degree on six of ten completed cells and zero on four, and both slopes are unchanged ($m = 4$: $6 \longrightarrow 7$ against $5 \longrightarrow 6$ from $t = 2$ to 3; $t = 3$: $6 \longrightarrow 7$ against $5 \longrightarrow 6$ from $m = 3$ to 4). The one-degree offset at $m \geq 3$ arrives with the degree-4 doubling links and the extra unknowns $u'_k$, a presentation shift, not a change in growth. Section 4 item 3 stays closed as far as this check can test it; not tested: [3]- and [4]-multiples together, larger $N$, an F4 run on the augmented systems.

## 7. Leads, ranked

1. Finish the decisive PKM cells: $t = 14$ cap 7; the $m = 4$, $N = 20$ closure (Amendment 8 readings).
2. The multiples presentation probe at $t = 2, 3$, $m = 2..5$ (minutes on a 4-core host).
3. The null-support slope: `null` control and supports of width $w = 1..4$; if $d_F(N, w)$ is curve-independent, the growth is a property of the carry-graded one-generator closure and the lane closes with a mechanism.
4. The first counted algebraic-track row: `pkm-semaev` on a Table B curve with the §5a band and matched rho (D-EKS approved; about 49 CPU-hours).
5. Generic-track constants (hash-table prefetch Design A; $N > 45$ at 20 bits): the only place a measured improvement over the current baseline is available.
6. Honest non-uniform accounting ($ST^2$ per row; preprocessing-rho reference).

What would change the list: an equation source that passes the filter of section 4. None is known; the jet exit is the only one of the three still open in the atlas.

## 8. Sources checked

Knowledge base unavailable (`research-kb` MCP absent; `scripts/research-kb status`: no executable); fallback: the lead atlas (`crypto-autoresearcher`, 176 leads, 30 in `ic-prime/`, `--match` run per idea) and the repository files listed in README §0. Problem record: `problems/prime-field-cross-summand-equation-budget.md`.